

AnveshakHub Ecosystem

Final Enterprise Specifications & Product Architecture Blueprint

This master specification blueprint details the end-to-end user journeys, functional requirements, relational models, security designs, and infrastructure deployment specifications for the AnveshakHub Ecosystem platform. The first release target is a Responsive Web Application designed utilizing Next.js, React, TypeScript, and Tailwind CSS, built to be fully PWA-ready for future mobile enablement without backend mutations.

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TARGET PLATFORM:

Responsive PWA Web (Next.js + React + TS) | NestJS REST API | PostgreSQL

VERSION STATUS:

v3.0 (Official Enterprise Release)

PART I: Stakeholder Analysis and User Journey

1. Purpose of the Document

The purpose of this document is to identify the stakeholders involved in the AnveshakHub ecosystem and define their journey within the proposed platform.

The platform involves multiple types of users, each having different responsibilities and permissions. Understanding their journey helps in designing a system that aligns with the actual workflow of AnveshakHub.

2. Stakeholders

The proposed platform consists of four primary stakeholders.

1. Industry
2. Expert
3. Student
4. AnveshakHub Administration

Each stakeholder interacts with the platform differently.

2.1 Complete Stakeholder Taxonomy

To satisfy complex enterprise projects, the platform implements a complete stakeholder taxonomy with distinct visibility, permission sets, and approval flows.

Industry Stakeholders

- Startups & MSMEs: Businesses seeking local funding, mentorship, and rapid prototyping.
- LLPs, Private & Public Limited Companies: Corporate entities proposing advanced R&D projects.
- Research & Government Organizations: Entities focusing on national-level research projects and public works.

Academic & Industry Experts

- Academic Experts: Professors, Associate Professors, and Assistant Professors directing university labs.
- Research Experts: Research Scientists, Post Docs, and PhD Scholars handling core technical deliverables.
- Industry Advisors: Technology Managers, Architects, Cyber Security Officers, Testing Officers, and Quality/Regulatory Experts.
- Contracted Advisors: Consultants, Freelancers, and External Contributors delivering specialized consulting modules.

Strategic Partners & Support Roles

- Government Organizations: Sponsoring grants, reviewing eligibility schemes, and tracking budget payouts.
- Research Labs & Facilities: Provisioning testing equipment, clean rooms, and computing clusters.
- Partners & Vendors: Supplying raw materials, software licensing, and specialized manufacturing equipment.

2.2 Stakeholder Responsibilities & Permissions Matrix

A. Academic Experts (Professors, Associate/Assistant Professors, Post Docs, PhDs)

- Responsibilities: Supervise technical direction, evaluate student performance, assign sprint tasks.
- Permissions: Create project milestones, request equipment and raw materials, access project repositories.
- Visibility: Hidden profile model; visible only to verified administrators and clients upon active match.
- Approval Flow: Verified via institutional verification systems, approved by AnveshakHub Admin.
- Contribution Model: Scientific guidance, project management, and final approval of deliverables.
- Future Scope: Automatic citation sync, automated co-authoring tools.

B. Industry Experts (Tech Managers, Architects, Cyber Security, QA, Regulatory)

- Responsibilities: Perform code audits, check regulatory compliance, enforce quality standards.
- Permissions: Access specific project code repos, audit logs, run security verification pipelines.
- Visibility: Profile visible to admins, project team leads, and corresponding clients.
- Approval Flow: Corporate domain verification, credentials validation, admin sign-off.
- Contribution Model: Security scorecards, peer reviews, architectural compliance reviews.
- Future Scope: Dynamic automated compliance reporting.

C. Consultants, Freelancers, & External Contributors

- Responsibilities: Deliver specific features, execute research modules, supply project consultation.
- Permissions: Write access to designated repositories, access to communication channels.
- Visibility: Visible to project team members during active contract periods.
- Approval Flow: ID check, NDA/MoU digital consent, and technical screening by administrative panel.
- Contribution Model: Direct software commits, technical documentation, scheduled consulting hours.
- Future Scope: Smart contract automated payouts based on task completion.

D. Government Partners, Research Labs, & Vendors

- Responsibilities: Sponsor grants, approve innovation schemes, lease facilities, deliver materials.
- Permissions: Upload available materials and equipment catalogs, track grant funding milestones.
- Visibility: Visible on marketplace search and grant discovery dashboards.
- Approval Flow: Institutional background check, entity documentation audit, MoU signing.
- Contribution Model: Provisioning physical resources, computational power, and funding.
- Future Scope: Unified facility booking calendar integration.

3. Industry User Journey**Step 1: Registration**

The industry creates an account by providing:

- Company Name
- Company Type
- Industry Domain
- Company Website
- Contact Person Details
- Official Email Address
- Contact Number
- Company Registration Documents

After registration, a company profile is created.

Step 2: Profile Completion

The industry completes its profile by providing:

- Company Description
- Services Offered
- Current Challenges
- Technology Requirements
- Preferred Solution Areas

Step 3: Problem Statement Submission

The industry submits a problem statement.

Information may include:

- Problem Title
- Problem Description
- Expected Outcome
- Preferred Timeline
- Required Expertise
- Supporting Documents

Step 4: Approval Process

After submission, the problem statement enters review. At this stage:

- Industry can submit requirements.
- Industry cannot directly approach experts.
- Expert access remains restricted.

The AnveshakHub team reviews the request.

Step 5: Discussion and Meetings

The AnveshakHub team may:

- Contact the industry
- Schedule meetings
- Understand requirements
- Evaluate feasibility

Multiple meetings may occur before approval.

Step 6: Client Activation

If approved:

- Industry becomes an active client.
- Expert access is enabled.
- Project workflow becomes available.

If rejected:

- Request remains closed.
- Industry can submit future requests.

Step 7: Project Monitoring

Once a project begins, the industry can:

- View project progress
- View milestones
- Attend meetings

- Upload documents
- Review deliverables

4. Expert User Journey

Experts may include:

- Professors
- Technical Professionals
- Consultants
- Researchers
- Domain Experts

Step 1: Registration

The expert creates a profile. Information includes:

- Name
- Expertise
- Experience
- Qualifications
- Certifications
- Publications
- Previous Projects

Step 2: Verification

The AnveshakHub team verifies the expert profile before activation. Only approved experts become active users.

Step 3: Project Discovery

Experts can only view projects that have been approved by the AnveshakHub team. Unapproved requests remain hidden.

Step 4: Project Acceptance

Experts can:

- Review project details
- Evaluate requirements
- Accept participation

Step 5: Team Creation

After accepting a project, experts can:

- Create internship opportunities
- Define student requirements
- Define team size

Example: Maximum Students = 30

Applications remain open until the limit is reached.

Step 6: Student Evaluation

Experts can:

- Review applications
- Evaluate resumes
- Accept candidates
- Reject candidates

If students leave or are removed, available slots automatically reopen.

Step 7: Project Execution

Experts guide:

- Students
- Technical Teams
- Research Activities

while tracking project progress.

5. Student User Journey

Step 1: Registration

Students create profiles. Information includes:

- Academic Information
- Skills
- Resume
- Certifications
- Project Experience

Step 2: Profile Building

Students continuously improve their profiles by adding:

- Skills
- Certifications
- Achievements
- Completed Projects

Step 3: Opportunity Search

Students can view available opportunities created by experts. Examples:

- Internships
- Research Projects
- Industry Projects

Step 4: Application

Students apply to opportunities by submitting:

- Resume
- Supporting Documents
- Profile Information

Step 5: Selection

Experts review applications and make selections. Selected students join project teams.

Step 6: Participation

Students contribute to:

- Development Activities
- Research Tasks
- Documentation
- Project Deliverables

under expert supervision.

6. AnveshakHub Administration Journey

The administration team controls the overall platform.

Responsibilities

User Verification

Approve:

- Industries
- Experts
- Students (if required)

CRM Management

Track:

- New Industries
- Meetings
- Discussions
- Approval Status

Project Governance

Monitor:

- Active Projects
- Experts
- Teams
- Progress

Platform Administration

Manage:

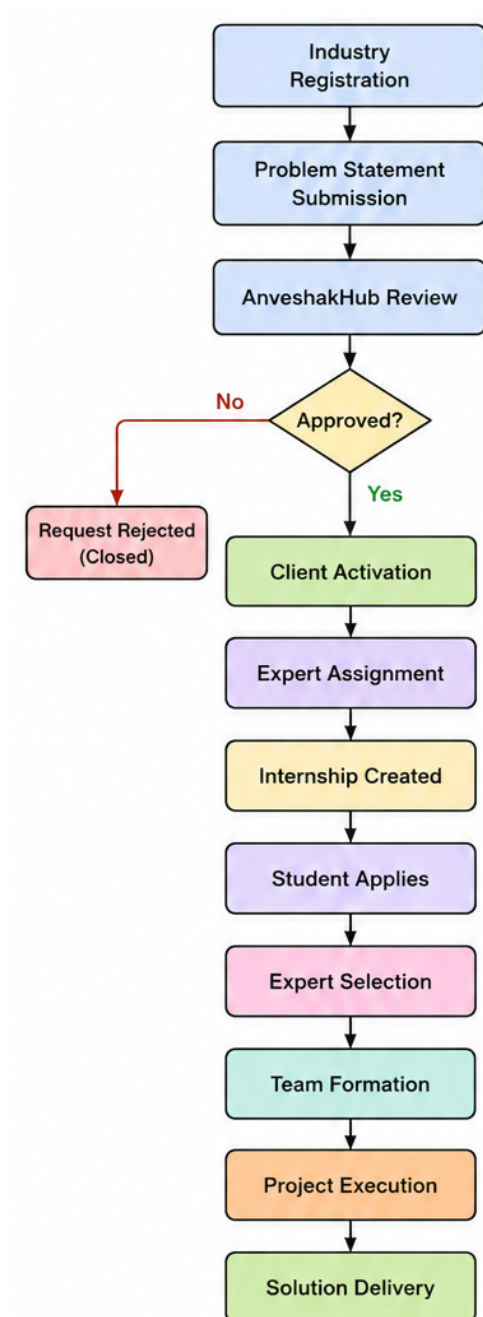
- User Accounts
- Permissions
- Reports
- Platform Operations

7. Summary

The platform is designed around four key stakeholders: industries, experts, students and the AnveshakHub administration team. Each stakeholder has a clearly defined journey and permission level. This structure helps ensure that project execution remains organized while allowing AnveshakHub to maintain control over approvals, expert engagement and overall project governance.

8. USER INTERACTION FLOW

The following flowchart represents the complete end-to-end user interaction and project lifecycle within the AnveshakHub Ecosystem.



This flow ensures a structured and transparent process from industry registration to final solution delivery through expert and student collaboration, governed by AnveshakHub.

PART II: Functional Requirements Specification (FRS)

1. System Components

The AnveshakHub Ecosystem consists of two primary system interfaces:

1.1 Responsive Web Application

Purpose & UI Stack:

Provide a centralized, responsive web platform utilizing Next.js, React, TypeScript, and Tailwind CSS. The system is Progressive Web App (PWA) ready, allowing future native wrappers without backend migrations.

Key Features:

- User Registration & Authentication
- Profile Management
- Project Participation
- Internship Management
- Notifications & Updates
- Document Upload & Access
- Responsive Navigation & Offline Sync Support

1.2 Web Administration Panel

Purpose:

Provide complete administrative control over platform operations and compliance verification.

Key Features:

- User Verification
- CRM Management
- Project Governance
- Meeting Management
- Analytics & Reporting
- Platform Monitoring

2. Authentication Module

Purpose:

Secure access management for all stakeholders.

Functional Requirements:

- User Registration
 - Industry Registration
 - Expert Registration
 - Student Registration
- Login Management
 - Email Authentication
 - Password Encryption
 - Session Management

- Password Recovery
 - Forgot Password
 - Reset Password
 - OTP/Email Verification
- Account Verification
 - Approve Users
 - Reject Users
 - Suspend Accounts
 - Reactivate Accounts

3. Industry Module

Purpose:

Enable industries to submit challenges and monitor project execution.

Functional Requirements:

A. Industry Dashboard

- Company Information
- Active Projects
- Project Status
- Notifications

B. Problem Statement Management

- Create Problem Statements
- Edit Problem Statements
- Upload Supporting Documents
- Track Approval Status

C. Meeting Management

- Request Meetings
- View Meeting Schedule
- Access Meeting Notes

D. Project Monitoring

- View Milestones
- Monitor Team Progress
- Review Deliverables
- Download Reports

4. Expert Module

Purpose:

Enable experts to guide projects and manage student teams.

Functional Requirements:

A. Profile Management

- Expertise Details
- Qualifications
- Certifications
- Previous Projects

B. Project Participation

- Browse Approved Projects
- Accept Assignments
- View Requirements
- Track Progress

C. Opportunity Management

- Create Internships
- Edit Opportunities
- Close Opportunities
- Manage Applications

D. Student Evaluation

- Resume Review
- Candidate Selection
- Candidate Rejection
- Team Formation

E. Project Monitoring

- Assign Tasks
- Update Milestones
- Submit Reports
- Track Performance

5. Student Module**Purpose:**

Enable students to participate in industry projects and internships.

Functional Requirements:**A. Profile Management**

- Resume Upload
- Skills Management
- Certifications
- Achievements

B. Opportunity Discovery

- Browse Internships
- Browse Research Projects
- Filter Opportunities
- View Deadlines

C. Application Management

- Submit Applications
- Upload Supporting Documents
- Track Application Status

D. Project Participation

- View Assigned Tasks
- Submit Deliverables
- Track Progress

- Communicate with Experts

6. Enterprise Verification Engine

Business Problem:

Manual entity and credentials check leads to slow onboarding, security threats, and documentation fraud.

Proposed Solution:

A unified verification engine that executes distinct automated trust workflows based on stakeholder type, integrated with digital signature verification.

Business Benefit:

Onboarding latency drops by 85%, security risk is mitigated, and legal auditability is established for all active engagements.

A. Stakeholder Workflows

- Startups & MSMEs: Automated validation via corporate registration registry and tax identity checks.
- LLPs, Private & Public Ltd: Direct document audits, director listing checks, and bank KYC verification.
- Government & Research Orgs: Authorized signing officer validation and official MoU check.
- Experts & External Contributors: Institutional domain checking, academic records OCR, and employment history audits.

B. Core Features

- Document OCR Engine: Automatic extraction of entity names, registration IDs, and dates from PDFs.
- Identity Verification: Biometric match against official credentials and government domain verification.
- Employment Check: Domain mail verification and digital signature matching.
- Legal Gate: Automated digital signing of NOC, NDA, MoU, and digital consent forms.
- Real-Time Tracking: User dashboard with verification status, pending documents, and audit logs.

C. Future Enhancement

Integrating decentralized block-chain credentials for instant verification.

7. Industry Engagement & CRM Module

Business Problem:

Manual client tracking leads to fragmented sales pipelines, lost requirements details, and slow proposal cycles.

Proposed Solution:

Integrated CRM system that registers new industry leads, schedules and logs discussions, manages proposal versions, and tracks client conversions.

Business Benefit:

30% increase in lead conversion, centralized discussion history, and transparent proposal tracking.

A. CRM Core Features

- Lead Management: Capturing incoming corporate queries and categorizing them by project scope.

- Meeting Lifecycle: Scheduling interface, automated invitations, action items, and minute generation.
- Discussion History: Audit trail of all communication, calls, email threads, and meeting transcripts.
- Proposal Tracking: Version-controlled drafts, client reviews, and approval pipelines.
- Client Conversion: State machine logic managing Lead -> Opportunity -> Active Client -> Archival.
- Health Dashboard: Analytics of client satisfaction, response times, active sprints, and renewal schedules.

B. Future Enhancement

Generative AI analysis of discussion history to automatically draft proposals.

8. Expert Resource Marketplace

Business Problem:

Research projects halt when experts lack equipment, software, or raw materials, leading to project delays.

Proposed Solution:

A unified resource marketplace where experts request physical, software, or advisory assets, managed and fulfilled by administration.

Business Benefit:

40% reduction in procurement overhead, optimized sharing of lab space, and transparent budget tracking.

A. Marketplace Request Categories

- Hardware/Equipment: Lab instruments, development boards, and computing hardware.
- Software Licenses: Specialized simulation tools, analytical frameworks, and IDEs.
- Raw Materials: Chemical supplies, metal stock, electronic components for physical prototypes.
- Facilities: Clean rooms, high-performance computing (HPC) clusters, and testing chambers.
- Advisory/Consulting: Access to peer experts, regulatory specialists, and legal counsels.

B. Administrative Control Workflow

- Request Generation: Expert selects category, quantities, reasons, and target project milestones.
- Feasibility Audit: System matches availability across internal inventory and lab calendars.
- Approval & Assignment: Admin panel assigns procurement paths, books lab slots, or denies requests with feedback.
- Vendor Lifecycle: System forwards orders to matched partners and records arrival status.

C. Future Enhancement

IoT sensors to automate door locks and equipment logging based on marketplace bookings.

9. Government Grants & Innovation Schemes

Business Problem:

Startups and MSMEs lack insight into government schemes, grants, and tax incentives, making capital lookup highly tedious.

Proposed Solution:

An algorithmic matching system linking active profiles with national grants, allowing unified application submission and auditing directly through AnveshakHub.

Business Benefit:

Enhanced non-dilutive funding, accelerated startup project rates, and simplified grant submission audits.

A. Module Features

- Discovery Index: Database of active government grants, tax credits, and support programs.
- Eligibility Matching: Automated analysis based on company domain, size, and R&D scope.
- Unified Submission: Wizard tool allowing application compiling, document attaching, and submission.
- Milestone Reporting: Financial tracker reporting matching progress back to government agencies.

B. Future Enhancement

API synchronization directly with government grant review portals.

10. Enterprise Universal Search

Business Problem:

Navigating cross-entity files, user profiles, and logs becomes slow as database records scale.

Proposed Solution:

Elasticsearch/OpenSearch index indexing user profiles, project metadata, requirements, documents, and logs with secure role-filtering.

Business Benefit:

Sub-second access to critical documents, reduced navigational clicks, and high resource discoverability.

A. Universal Search Architecture

- Autocomplete Suggestions: Dynamic prefix matching displaying relevant files, groups, or projects.
- Secure Filtering: Facets filtering search outputs by role access, tags, dates, and verification state.
- Quick Action Commands: System triggers allowing direct actions (e.g. '/verify' or '/call') via search.
- Saved Queries: User bookmarking, transaction histories, and query history storage.

B. Future Enhancement

Natural language search parsing context and returning relevant entities.

11. Confidentiality & Isolation Management

Business Problem:

Direct matching before contract signature risk bypass of AnveshakHub, intellectual property theft, and corporate leaks.

Proposed Solution:

Complete logic isolation between experts and corporate clients. Direct messaging is disabled until the NDA and project MoU are digitally executed.

Business Benefit:

Guarantees IP protection, secure platform compliance, and enforces platform retention.

A. Isolation Features

- Hidden Profile Architecture: Expert profiles are searchable by matching parameters but retain anonymous identity.
- Strict Access Controls: Messaging, phone lines, and code repository links remain locked until project kickoff.
- NDA Gatekeeper: Automated state machine unlocking channels only when matching digital NDAs/MoUs are verified.

B. Future Enhancement

Automatic document watermarking for shared review files.

12. Smart Registration & Dynamic Profiling

Business Problem:

Long registration forms lead to high drop-offs, inaccurate data entry, and incomplete user profiles.

Proposed Solution:

Dynamic smart forms using conditional logic, progressive profiling (requesting data only when needed), document OCR ingestion, and government API connections.

Business Benefit:

Onboarding drop-off falls by 70%, data entry is error-free, and profile quality is significantly higher.

A. Smart Registration Features

- Dynamic Conditional Forms: Inputs adapt based on selected role and registration category.
- Progressive Profiling: Initial registration asks for 3 fields; additional inputs are requested as the user performs actions.
- Document Ingestion: Drag-and-drop registration certificates where details are parsed by OCR and populated.
- API Integration: Instant lookup of company records and corporate registration IDs via API.

B. Future Enhancement

Direct verification integration with professional portfolio databases.

13. Enterprise Core Modules

The following modules form the core administrative and operations backbone of AnveshakHub:

CRM Module

- Purpose & Workflow: Track leads, schedules discussions, logs meetings, manages proposal workflows, and client health.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

Finance Module

- Purpose & Workflow: Invoicing, milestone payouts, corporate R&D funding allocations, taxation, and student stipends.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

HR Module

- Purpose & Workflow: Contract creation, expert credentials lookup, student team assemblies, and advisory time allocations.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

Operations Module

- Purpose & Workflow: Project sprint calendars, milestones validation workflows, task routing, and lab logs.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

Procurement Module

- Purpose & Workflow: Orders tracking, raw material requests matching, and software licensing dispatch.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

Legal Module

- Purpose & Workflow: NDA versions, MoUs, digital signature vault, patent requests tracking, and confidentiality auditing.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

Knowledge Repository

- Purpose & Workflow: Secure file repository storing research documentation, code libraries, and final review files.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

Communication Module

- Purpose & Workflow: Secure internal chat, notifications delivery, email alerts, and broadcast engines.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

Analytics Module

- Purpose & Workflow: Metrics tracking platform performance, project velocity, active expert ratios, and SLA tracking.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

Notification Module

- Purpose & Workflow: System alert engine dispatching push, mail, and message updates.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

Reports Module

- Purpose & Workflow: Financial statements compiling, project logs reporting, and corporate audit trail generation.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

Audit Module

- Purpose & Workflow: Cryptographic hash audit logs tracking all files access, permissions shifts, and document changes.
- Role Access: Fenced by role authorizations (Admin read-write, Experts/Clients conditional access).
- Future Enhancements: Integration with corporate accounting software.

14. Complete Project Lifecycle Flow

The platform models the complete end-to-end lifecycle of an industrial project through twenty distinct transition phases:

- 1. Lead Generation: Corporate contact is established, industry registers on the web platform, and profile is created.
- 2. Discussion / Meeting: First meeting scheduled via CRM to discuss raw ideas, goals, and outline potential collaborations.
- 3. Requirement Gathering: Formal capturing of requirements, functional definitions, expected outcomes, and budget constraints.
- 4. Proposal Drafting: Technical experts or admin draft initial solution architecture proposals and scope of work.
- 5. NDA Execution: Enforcement of NDA signing via legal gate before sharing detailed expert or industry IP.
- 6. MoU Signing: Drafting and signing of official MoU outlining milestones, resource allocations, and IP shares.
- 7. Admin Approval: AnveshakHub Administration audits documentation and activates the corporate project space.
- 8. Expert Assignment: Matching algorithm suggests top academic/industry experts; lead expert is assigned and matched.
- 9. Opportunity Creation: Lead expert outlines student internship opportunities, defining required skills and slots.
- 10. Student Applications: Students browse open opportunities, submit resumes, and apply through their portal.
- 11. Student Selection: Expert reviews profiles, runs evaluations, accepts qualified candidates, and closes vacant slots.
- 12. Team Assembly: Student and expert profiles are linked inside the project sandbox workspace, and workspace is initialized.
- 13. Sprint Development: Weekly developmental tasks are allocated, and code commits/deliverables are submitted.
- 14. Milestone Auditing: Progress validation against agreed timelines; QA experts review design parameters.
- 15. Security Testing: API security reviews, static code analysis, and role-based isolation gates verification.
- 16. User Acceptance Validation: Client logs in to dashboard, runs code packages, and accepts milestone outputs.

- 17. System Deployment: Automatic deployment pipelines build code packages and host on test/staging servers.
- 18. Knowledge Transfer: Compiling final documentation, research reports, and delivering repos to the client.
- 19. Project Maintenance: SLA support, minor patching, and performance tracking during the initial operational period.
- 20. Archival & Audit: Milestone transactions closed, files compressed to cold storage, and audit logs hashed.

15. Role & Permission Matrix

The following matrix details access configurations for all core stakeholder actions:

Role	View	Create/Edit	Approve	Delete	Download	Assign/Manage
Admin	All Data	All Entities	All Projects	All Entities	All Files	All Teams
Industry	Own Data	Own Challenge	MoU/Milestone	No Access	Assigned Files	No Access
Expert	Own Scope	Internships	Students	No Access	Syllabus/CVs	Student Tasks
Ext Contrib	Assigned Scope	Task Comments	No Access	No Access	Allowed Files	No Access
Student	Active Project	Own Profile	No Access	No Access	Study Materials	No Access
Government	Grants/Metrics	Grants/Schemes	Grant Funding	No Access	Audit Reports	Funding Payout
Partner	Own Orders	Catalog Details	No Access	No Access	Po/Orders Docs	Material Delivery
Vendor	Own Orders	Catalog Details	No Access	No Access	Invoice Docs	Material Delivery

16. Non-Functional Requirements (NFR)

- Performance: Sub-second API response time for 95% of requests. Elasticsearch global search matches returned in < 500ms.
- Scalability: Horizontally scale NestJS application layers and Next.js static files. Database read replication supports 10k concurrent sessions.
- Availability: Minimum 99.9% uptime target. Multi-availability zone database deployment with automated failovers.
- Reliability: RabbitMQ manages transaction queues to prevent message loss during API spikes.
- Maintainability: Modular Next.js app directory structure and microservice backend paths audited via sonar tools.
- Accessibility: Frontend pages fully optimized to WCAG 2.1 AA parameters, offering screen reader compatibility.
- Security: Periodic third-party security scans, automated JWT refresh tokens, and rate limits.
- Usability: Sleek shadcn/ui dashboard widgets with quick actions matching corporate workflows.
- Disaster Recovery: DB backup files stored on remote object repositories. Database RTO < 2 hrs, RPO < 15 mins.
- Business Continuity: High availability configuration for reverse proxies, database clustering, and cache clusters.

17. API Standards & Documentation

- Protocol Structure: JSON-based REST APIs utilizing standard HTTP verbs (GET, POST, PUT, DELETE, PATCH).
- Authentication: Auth token passed via Authorization: Bearer JWT header. Session refresh runs over secure, HttpOnly cookies.
- Versioning: URL versioning structure (e.g., /api/v1/projects) ensures seamless updates.
- Rate Limiting: Rate limit boundaries (100 req/min per IP) handled at the API gateway utilizing Redis clusters.
- Pagination Pattern: Cursor-based pagination on large listings prevents API response memory overhead.
- Error Handling: Unified JSON error responses following RFC 7807 problem details specifications.

- Swagger Documentation: Real-time Swagger endpoints auto-generated from NestJS decorators.
- Request Validation: Strict request validation schemas (class-validator) sanitizing inputs before execution.

18. Secure File & Document Management

- Secure Storage: Resumes, certifications, NDAs, and MoUs stored in MinIO object storage bucket.
- Access Permissions: Temporary pre-signed URL matching (valid for 15 minutes) generated for verified users.
- Deception Isolation: Projects files isolated and encrypted at-rest using customer-managed KMS keys.
- Virus Scanning: Upload files routed through inline antivirus scanning before committing to buckets.
- Version Control: Relational tracking of NDA and MoU document versions with audit hash footprints.
- Retention Policy: Closed project files compressed, hashed, and moved to cold archiving buckets after 5 years.

19. AI-Powered Services

- Expert Matching: Collaborative filtering models matching academic publications with industry problem domains.
- Student Recruitment: Parsing CVs against opportunity skill tags to rank qualified applicants.
- Project Recommendation: Suggesting funding schemes and corporate grants based on industry profiles.
- Resume Parsing: Automated NLP extraction of education, certifications, and skills from PDF files.
- Meeting Summarization: Off-line pipelines summarizing scheduled CRM discussions into action items.
- Skill Gap Analysis: Evaluating student performance histories against market needs to suggest training tracks.
- Smart Search: Semantic search parsing natural query terms and returning relevant database matches.

20. Analytics & Stakeholder Dashboards

- Admin Panel Dashboard: Aggregates total active users, active projects, SLA tracking, and audit alerts.
- Industry Dashboard: Displays milestones velocity, task statuses, meeting logs, and proposal versions.
- Expert Dashboard: Displays academic hours tracking, assigned student metrics, and resource requests.
- Student Dashboard: Displays internship matching scorecards, task lists, and cert rewards.
- Operations Dashboard: Real-time graphs showing CRM lead conversion steps, grant application status, and lab bookings.

21. Enterprise Audit Trail

- Activity Logging: Logging all user logins, document view actions, and entity updates.
- Audit Database: Audit records stored in partitioned log tables, capturing before/after JSON states.
- IP Tracking: Logging IP addresses, browser footprint headers, and timestamp footprints for

critical actions.

- Immutability: Audit log indices are write-once-read-many (WORM) and replicated to cold archive storage daily.
- Rollback Controls: Version control tracking enables rolling back project workspace states to prior active versions.

22. Workflow Engine

- Approval Rules: Configurable multi-signature validations for projects (Industry signoff -> Expert signoff -> Admin verify).
- State Transitions: Automated state updates (e.g., Draft -> Under Review -> Active -> Completed) triggered by action items.
- Escalation Logic: Automated email/SMS notifications dispatched to admins if milestone reviews delay past 72 hours.
- Task Allocation: Smart routing of task slots based on expert assignment rules and student capacity limits.

23. External Integrations (Future Ready)

- Google Calendar & Meet: Automated calendar booking and meeting link generation for CRM discussions.
- Microsoft Teams: Channel webhooks dispatching milestone updates to client communication spaces.
- GitHub Integration: Direct repository creation, user assignment, and commit activity tracking inside sandboxes.
- LinkedIn: Auto-publishing credentials verification cards upon internship completion.
- Govt API Registry: Verification hooks to business registry databases and academic certification portals.
- Email & SMS: Dispatching system updates via Resend/SendGrid and SMS gateways.

24. Project Risk Register

The following matrix captures initial risks, impacts, and mitigation plans:

Risk Description	Likelihood	Impact	Mitigation Strategy	Owner
Expert-Industry bypass	Medium	High	Anonymous profiles prior to MoU signing	Legal Admin
Corporate IP leakage	Low	Critical	Automated NDAs & sandbox environments	Security
Verification process delay	Medium	Medium	OCR parsing and Govt API integration	Operations
Student high attrition rate	High	Medium	Standby candidate pools and standby queues	Expert Lead
Service gateway downtime	Low	High	Redis caches, replicas & Nginx rules	DevOps

25. Document Version History

The history of modifications made to this system specification document:

Version	Date	Author	Changes Summary
v1.0	June 2026	Vishnu SH & Pranav SP	Initial specification and user journey drafting.
v2.0	July 2026	Pranav SP	Added Verification Engine, CRM details, and initial architecture.
v3.0	July 2026	Pranav SP	PWA responsive upgrade, integrations & deployment.

26. Assumptions & Constraints

- Business Assumptions: Stakeholders have active internet connections; academic credentials can be validated online.
- Technical Assumptions: MinIO buckets are fully accessible; Next.js frontend has direct connection path to Nginx proxy.
- Known Constraints: Initial API verifications are bound to response limits of governmental databases.
- Out of Scope: Multi-tenant SaaS setups for external corporate entities; custom offline document signing systems.

PART III: System & Technology Architecture

1. Architecture Overview

The proposed platform utilizes a centralized, secure architecture where responsive Next.js web clients communicate with NestJS backend services. The components are coordinated as follows:

- Next.js Frontend: Serves responsive web interface, optimized using Tailwind CSS & TypeScript (PWA ready).
- Nginx Proxy: Gateway proxy executing SSL/TLS termination, request routing, and client caching.
- NestJS API: Core backend service routing database calls, authorization guards, and background workflows.
- Redis Cache: Fast retrieval layer for active sessions, security rate limits, and configuration caches.
- RabbitMQ Queue: Message broker managing asynchronous task dispatches (e.g. notifications, OCR processes).
- PostgreSQL Database: Relational storage engine managing user models, matching rules, and transaction states.
- MinIO Object Storage: S3 compatible file repository housing verification PDFs, NDAs, MoUs, and resumes.
- OpenSearch Index: Distributed search engine tracking profiles, requirements metadata, and system logs.
- Monitoring Stack: Centralized logging and metrics visualization via Prometheus, Loki, and Grafana dashboards.

2. Deployment Architecture

- Containerization: All services packaged into separate Docker containers, managed via Docker Compose in staging.
- Reverse Proxy: Nginx manages ingress traffic, balancing loads across active application containers.
- Application Server: NestJS node instances executed in cluster mode, ensuring performance scales with server cores.
- Database Server: PostgreSQL hosted on dedicated volumes with active-passive read replica configurations.
- Object Storage: MinIO hosted on clustered disks, backed up to off-site cloud archives daily.
- Messaging & Caching: RabbitMQ and Redis deployed in clustered setups with persistent volume maps.
- Monitoring Nodes: Prometheus, Loki, and Grafana collect system statistics, generating automated alerts.
- Backup & Recovery: Automated snapshot runs daily. Cold archiving storage secures transactions history.
- Future Cloud Migration: Mapped paths to transition services to fully managed AWS / Azure environments.

3. Tech Stack Selection Summary

Component	Technology Chosen
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Frontend Client	Next.js, React, TS, Tailwind CSS, shadcn/ui (PWA Ready)
Web Administration Panel	React.js with TypeScript
Backend API Services	NestJS (TypeScript)
Database Layer	PostgreSQL with Prisma ORM
Caching & Limits	Redis Cache Cluster
Messaging & Task Queue	RabbitMQ
File Storage Layer	MinIO Object Storage (S3 API compatible)
Enterprise Search Index	OpenSearch
Containerization & Proxy	Docker & Nginx Reverse Proxy
Monitoring Services	Prometheus, Loki & Grafana Stack
CI/CD & Version Control	Git, GitHub & GitHub Actions
Testing Tools	Jest (Backend/Unit), Playwright (E2E), Postman

4. Security Architecture

- Authentication: Secure OAuth2 flow. Access tokens valid for 15 minutes; refresh tokens stored in secure, HttpOnly, SameSite cookies with a 7-day expiry.
- Authorization: Strict RBAC (Role-Based Access Control) with resource access guards. Transition path mapped to dynamic ABAC (Attribute-Based Access Control) for contextual contract evaluations.
- Gateway Protections: API Gateway managing rate limits (Redis-backed token bucket, max 100 req/min/user), CORS constraints, and TLS termination.
- Data Encryption: AES-256 for database fields at rest. TLS 1.3 for in-transit network packets. Passwords hashed using bcrypt (cost factor 12).
- Secrets Management: Encryption keys, credentials, and API tokens stored and rotated via automated secrets manager.
- Backup & Disaster Recovery: Daily geo-replicated snapshots. Target RTO (Recovery Time Objective) < 2 hours, RPO (Recovery Point Objective) < 15 minutes.
- Compliance Policy: Enforced audit trail logging, data minimization matching GDPR, and document retention policies.

5. Database Architecture & Relational Design

- Core Database Entities: Users, Profiles, ProblemStatements, Projects, Internships, Applications, Meetings, VerificationDocuments, Resources, Grants, AuditLogs.
- Entity Relationships: Users 1:1 Profiles. Profiles 1:N ProblemStatements. ProblemStatements 1:1 Projects. Projects 1:N Internships/Meetings/Resources. Internships 1:N Applications.
- Indexing Strategy: B-Tree index on Users(email), Profiles(user_id), and Applications(internship_id). GIN index on ProblemStatements(title, description) for full-text search.
- Storage Estimation: Initial active database size estimated at 500 MB. Daily logs, search indices, and activity audit logs will grow at ~1.5 GB per year.
- Archival Policy: Partitions of AuditLogs and transactions older than 365 days are compiled and shifted to cold storage.
- Audit & Log Shadow Tables: System shadow tables capture before/after states of critical updates (financials, project statuses, permission sets).
- Asset Storage: Binary documents (resumes, NDA/MoU PDFs, corporate registration certificates)

stored on AWS S3 with IAM restricted lifecycle policies.

6. Enterprise UI/UX Design Guidelines

- Minimal Layout: Slate grey backgrounds, deep navy branding highlights, structured grids, and clean visual cards.
- Role Dashboards: Separate views and dashboard analytics metrics tailored to active stakeholder roles.
- Global Action Hub: Central command bar allowing instant navigation, matching, and shortcuts.
- Accessibility: Text contrast ratios and page structures compliant with WCAG 2.1 AA parameters.
- Responsive Design: Responsive fluid layouts adapting to mobile screens (Flutter) and desktop interfaces (React).

7. Quality Assurance & Testing Strategy

- Unit Testing: Automated test coverage on models and operations classes (NestJS/React) targeting > 85% lines coverage.
- Integration Testing: Verification of database query logic, transaction rollbacks, and schema migrations.
- API Testing: Validation of status codes, parameter checking, and error models on REST endpoints.
- System & E2E Testing: Automated browser actions executing critical registration, verification, and payment flows.
- Acceptance & Regression Testing: End-user acceptance checks and automated CI/CD validation cycles.
- Release Quality Checklist: Code review, security scan, DB migration check, deployment validation, audit configuration verification.

8. Future Architecture Enhancements

- AI recommendation modules to suggest matching experts for problem statements.
- Automated transcription and summaries for scheduled CRM discussion meetings.
- Intelligent project scoring metrics to track team execution performance.
- Integration endpoints to external enterprise software suites.

This complete specification serves as the conceptual blueprint for database schema design, REST API development, UI/UX mockups, and mobile client implementation.

--- End of Specifications ---